

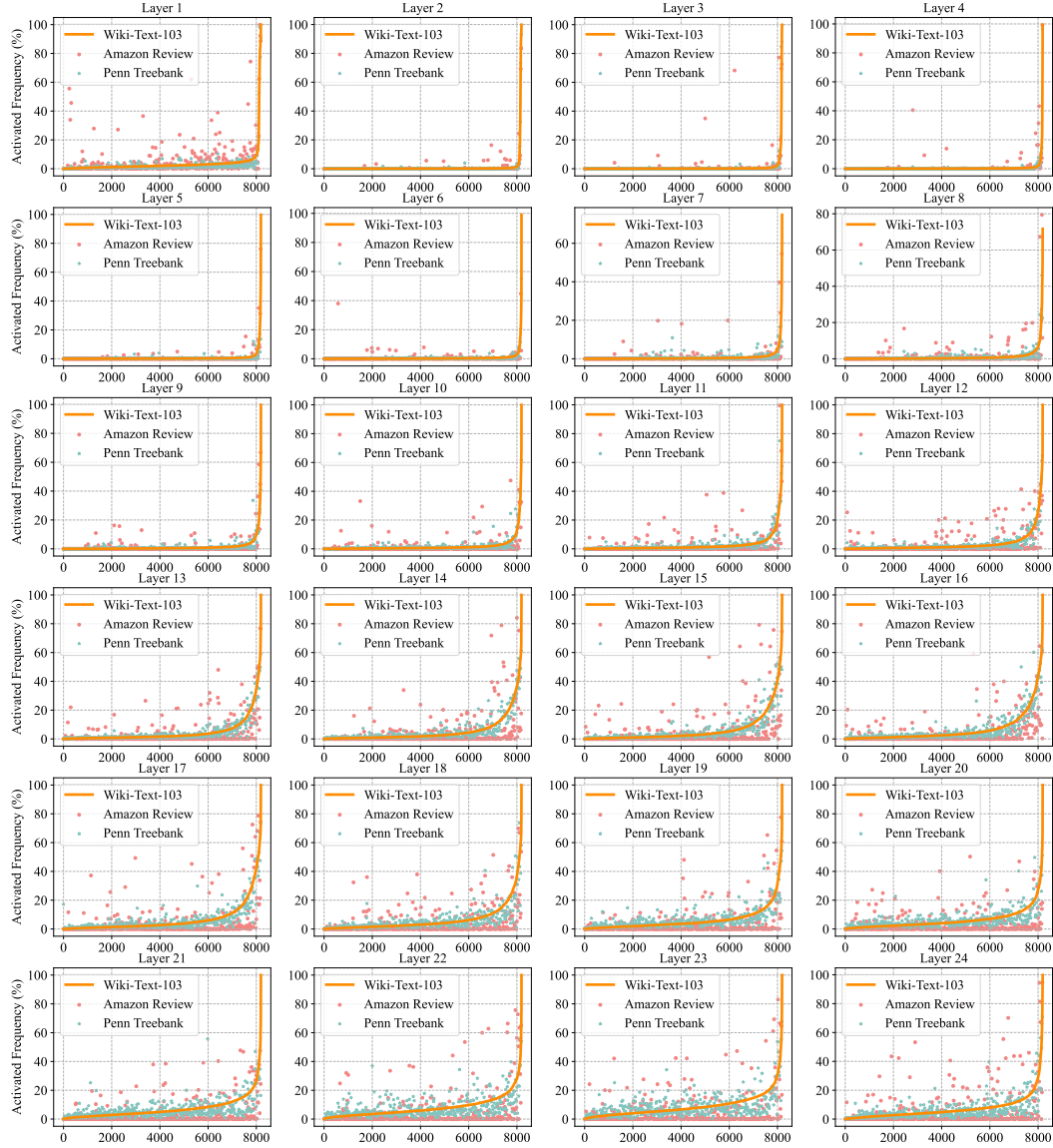


Figure 12: The distribution of activated frequency across diverse input content. The x-axis represents the index of neurons, which is ordered by the activated frequency from Wiki-Text-103.

D Theoretical Analysis

Recently, a number of works have studied the attention scheme in LLMs from a theoretical perspective [96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122]. In this work, we provide a different and novel angle compared to the previous work. We present the concept of the submodular property and propose an eviction policy, known as greedy H_2 , which is a modification of dynamic submodular maximization. Furthermore, assuming the attention scheme to be submodular, we establish that constructing the set S_i without any cache size limitation satisfies the near-optimal property in terms of submodularity. We provide theoretical guarantees for our robust and approximate greedy eviction policy algorithm (Algorithm 2). Due to space limitation, we only give informal description of algorithm (Algorithm 2) in Section 4.1. In Section D.6, we give an algorithm (Algorithm 2) which has full and complete implementation details for Algorithm 1. We also offer a mathematical formulation for sparsity preservation that is observed in Section 3 and proposed an algorithm (Algorithm 4) to solve the problem.

Specifically, in Section D.1, we provide several basic definitions and notations. In Section D.2, we briefly the definition of submodular function. In Section D.3, we define the dynamic submodular framework, which gives the formal version of Definition 4.1. In Section D.4, we briefly review the static attention computation problem. In Section D.5, we formulate the attention computation in recursive fashion. In Section D.6, we briefly review our eviction policy, which gives the formal version of Definition 4.3. In Section D.7, we discuss the diminishing return for submodular. In Section D.8, we discuss the high-level ideas for submodular. In Section D.9, we analyze the robust greedy algorithm error propagation. In Section D.10, we explain how to add items into sets via approximate function. In Section D.11, we provide several definitions related to dynamic properties. In Section D.12, we prove an induction lemma for the exact function. In Section D.13, we prove an induction lemma for the approximation function. In Section D.14, we provide theoretical guarantees for both the full-knowledge version (formal version of Lemma 3.1) and the limited-cache-size version (formal version of Theorem 4.4). In Section D.15, we provide a more detailed discussion of theoretical work about attention computation and regression-related problems. In Section D.16, we provide a mathematical formulation for sparsity preserving. In Section D.17, we provide the definition of loss function which can potentially generate sparse (heavy hitter type attention sore). In Section D.18, we explain how to compute the gradient of the loss function. In Section D.19, we show how to compute the Hessian of the loss function. In Section D.20, we show that Hessian is positive definite. In Section D.21, we prove the Lipschitz property for the Hessian matrix. In Section D.22, we show that using a gradient-type algorithm is sufficient to optimize that (heavy hitter type) loss function.

D.1 Notations

For a positive integer n , let $[n] := \{1, 2, \dots, n\}$.

For a vector $x \in \mathbb{R}^n$, let $\sqrt{x} \in \mathbb{R}^n$ denote the vector with the i -th entry being $\sqrt{x_i}$ and $\text{diag}(x) \in \mathbb{R}^{n \times n}$ denote the diagonal matrix with the i -th diagonal entry being x_i . For two matrices $A, W \in \mathbb{R}^{n \times n}$, let $\|A\|_W := (\sum_{i=1}^n \sum_{j=1}^n W_{i,j} A_{i,j}^2)^{1/2}$ and $W \circ A$ denote the matrix where $(W \circ A)_{i,j} = W_{i,j} A_{i,j}$. For matrix $W \in \mathbb{R}^{n \times n}$, let $D_{W_i} := \text{diag}(W_{i,:})$ with $i \in [n]$.

For two vectors $x \in \mathbb{R}^n$ and $w \in \mathbb{R}_{\geq 0}^n$, let $\|x\|_w := (\sum_{i=1}^n w_i x_i^2)^{1/2}$. For a vector x , its ℓ_2 norm is defined as $\|x\|_2 := (\sum_{i=1}^n x_i^2)^{1/2}$ and its ℓ_p norm is defined as $\|x\|_p := (\sum_{i=1}^n |x_i|^p)^{1/p}$. For a square matrix A , we denote $\text{tr}[A]$ as the trace of matrix A .

For a matrix $A \in \mathbb{R}^{n \times k}$ (suppose $n \geq k$), we use $\|A\|$ to denote its spectral norm, i.e., $\|A\| = \sup_x \|Ax\|_2 / \|x\|_2$. We use $\|A\|_F$ to denote its Frobenius norm $\|A\|_F := (\sum_{i=1}^n \sum_{j=1}^k A_{i,j}^2)^{1/2}$.

Suppose matrix $A \in \mathbb{R}^{n \times k}$ has SVD decomposition $U\Sigma V^\top$ where $U \in \mathbb{R}^{n \times k}$ (this matrix has orthonormal columns), $\Sigma \in \mathbb{R}^{k \times k}$ is a diagonal matrix, and $V \in \mathbb{R}^{k \times k}$. We call columns of U singular vectors. We use $A^\dagger \in \mathbb{R}^{k \times n}$ to denote the Moore-Penrose pseudoinverse, then $A^\dagger = V\Sigma^{-1}U^\top$. Suppose $\Sigma \in \mathbb{R}^{k \times k}$ is sorted diagonal matrix, let $\sigma_1, \dots, \sigma_k$ denote the diagonal entries of Σ . Then we call σ_i the i -th singular value of the matrix, and we write it as $\sigma_i(A)$.

For any symmetric matrix $B \in \mathbb{R}^{k \times k}$, we denote its eigenvalue decomposition as $U\Lambda U^\top$, where Λ is a diagonal matrix. Let $\lambda_1, \dots, \lambda_k$ denote the entries on diagonal of $\Lambda \in \mathbb{R}^{k \times k}$. We say λ_i is the i -th eigenvalue. Usually, we write it as $\lambda_i(B)$.

The connection between eigenvalues and singular values is

$$\sigma_i^2(A) = \lambda_i(A^\top A)$$

We use the notation $A \succeq 0$ to denote that matrix A is positive semidefinite (psd). Mathematically, $A \succeq 0$ means for all vectors x , we have $x^\top Ax \geq 0$.

Similarly, for two squarer matrices A and B , we use $A \succeq B$ to denote the case where for all vectors x , $x^\top Ax \geq x^\top Bx$.

Let $\Pr[\cdot]$ and $\mathbb{E}[\cdot]$ denote the probability and expectation. We define the maximum between a and b as $\max\{a, b\}$. We denote $\min\{a, b\}$ (resp. $\max\{a, b\}$) as the minimum (reps. maximum) between a and b .

Throughout, for non-negative real numbers a and b , we use the notation $a = (1 \pm \epsilon)b$ if $a \in [(1 - \epsilon)b, (1 + \epsilon)b]$.

D.2 Submodular

We provide the standard definition of the submodular function.

Definition D.1 (Submodular function [123]). *For a finite set Ω , a submodular function is defined as $f : 2^\Omega \rightarrow \mathbb{R}$, where 2^Ω denotes the power set of Ω . It is characterized by the fulfillment of any of the following equivalent criteria:*

- *Condition 1.*
 - For $S, T \subseteq \Omega$ with $S \subseteq T$ and every $a \in \Omega \setminus T$ we have that $f(S \cup \{a\}) - f(S) \geq f(T \cup \{a\}) - f(T)$
- *Condition 2.*
 - For every $S, T \subseteq \Omega$ we have that $f(S) + f(T) \geq f(S \cup T) + f(S \cap T)$.
- *Condition 3.*
 - For every $S \subseteq \Omega$ and $a_1, a_2 \in \Omega \setminus S$ such that $a_1 \neq a_2$ we have that $f(S \cup \{a_1\}) + f(S \cup \{a_2\}) \geq f(S \cup \{a_1, a_2\}) + f(S)$.

For convenience of discussion, in this paper, we always choose $\Omega = [n]$ when we want to discuss the submodular function.

Next, we provide some examples/types of submodular functions. One important class is called monotone,

Definition D.2 (Monotone). *A set function f is monotone if for every $T \subseteq S$ we have $f(T) \leq f(S)$.*

Here are a number of monotone submodular functions

- Linear (Modular) functions
 - A linear function can be represented as $f(S) = \sum_{i \in S} w_i$. If all the weights w_i are nonnegative, then the function f is considered monotone.
- Budget-additive functions
 - A budget-additive function has the form $f(S) = \min(B, \sum_{i \in S} w_i)$ where each weight w_i and the budget B are nonnegative.
- Coverage functions
 - We have a set $\Omega = \{E_1, E_2, \dots, E_n\}$ where each E_i is a subset of a broader set Ω' . The coverage function can be expressed as $f(S) = |\cup_{E_i \in S} E_i|$ for any $S \subseteq \Omega$. This function can be generalized by assigning non-negative weights to the elements.